BUSINESS REQUIREMENTS DOCUMENT

Blood Bank Management Module

Portfolio Case Study | Healthcare Information System

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1. Executive Summary

The Blood Bank Management Module is designed to streamline and manage the key activities involved in blood bank operations within a healthcare organization. The module brings together donor management, blood donation, immunohematology, blood component preparation, inventory management, quality assurance, patient blood requests, blood issue, and disposal of blood units into a single, structured workflow.

The main objective is to ensure that blood and blood components are properly tracked throughout their life-cycle, from donation and processing through storage, quality checks, issue, and final disposal. By maintaining accurate and up-to-date information, the system helps blood bank staff quickly identify available stock, process patient requirements, and manage blood units in accordance with defined procedures.

A key focus of the system is traceability. Important activities and status changes are recorded against the respective users and transactions, allowing staff to track the history of blood units and maintain an auditable record of operational activities. This provides better control over inventory, supports quality and compliance processes, and helps hospital stakeholders make timely decisions based on reliable blood bank information.

1.1 Business Value:

- Single source of truth for blood inventory and operational transactions.
- Faster identification of available blood units and request status.
- Reduced duplicate data entry and manual tracking.
- Improved traceability of donor, donation, unit and issue records.
- Improved management visibility through dashboards and reports.
- Foundation for future alerts, donor engagement and analytics.

1.2 Document Purpose:

This BRD defines the business requirements, scope, workflows, rules, data needs and acceptance expectations for the module. It is designed to demonstrate practical Business Analyst capabilities including requirements elicitation, process analysis, stakeholder analysis, functional decomposition, business rules, acceptance criteria and requirements traceability.

2. Business Context & Problem Statement

2.1 Business Context:

A Hospital blood bank must coordinate multiple related activities: donor registration, donation recording, blood-unit life-cycle management, inventory monitoring, patient requests from IP / OP/ ER/ Day Care / OT, approval and issue for these request from inter departments. These activities depend on accurate data and timely decisions.

2.2 Problem Statement

Without a structured digital workflow, blood bank operations may rely on fragmented records, manual checks or disconnected processes. This can make it difficult to determine current inventory, track the status of requests, identify units approaching expiry, or demonstrate an end-to-end history of a blood unit and Component.

2.3 Business Pain Points

- Difficulty maintaining a consistent view of available inventory.
- Manual effort in searching donor and patient information.
- Limited visibility into pending, approved, rejected and fulfilled requests.
- Risk of issuing a unit whose status does not permit issue.
- Limited audit-ability of changes to important records.
- Delayed identification of low-stock or near-expiry conditions.
- Reporting effort when management needs operational information.

2.4 Opportunity

A workflow-driven module can connect operational activities while preserving clear ownership, statuses and validation points. The solution should support staff rather than add unnecessary complexity.

3. Objectives & Success Measures

Objective	Success Measure
Improve inventory visibility	Users can view inventory by blood group, component and status.
Improve request turnaround	Request life-cycle is visible from creation through fulfillment.
Improve data quality	Mandatory fields and validation prevent incomplete core records.
Improve traceability	Key transactions retain timestamps, users and status history.
Reduce manual effort	Common donor, inventory and request searches are completed within the application.
Improve management visibility	Standard reports and dashboard indicators are available.

3.1 Key Success Indicators

- Inventory records accurately reflect the current application status of units.
- Users can identify pending requests without relying on separate trackers.
- Critical transactions are attributable to an authenticated user.
- Expired or otherwise non-eligible units are prevented from being issued.
- Business users can obtain standard operational reports without manual data consolidation.

3.2 Guiding Principles

- Patient safety and data integrity take priority over convenience.
- Least-privilege access should be applied.
- Important actions should be traceable.
- Statuses should be explicit and mutually understandable.
- Validation should happen as early as reasonably possible.

4. Stakeholders

Stakeholder	Role / Interest	Key Needs
Blood Bank Administrator	Owns configuration and operational oversight	Inventory visibility, reports, user controls
Blood Bank Technician	Performs day-to-day transactions	Fast donor, donation and inventory workflows
Doctor / Authorized Clinician	Raises or reviews patient requirements	Accurate request status and availability
Nurse / Hospital Staff	Coordinates patient-related requests	Request progress and issue information
Patient	Beneficiary of blood service	Timely fulfillment; appropriate communication through hospital process
System Administrator	Technical administration	Access control, availability, audit support
Hospital Management	Operational governance	KPIs, trends and exception reporting
QA / Testing Team	Validates requirements and behaviour	Testable acceptance criteria and traceability

5. Scope

5.1 In Scope

- Donor registration and donor profile maintenance.
- Donation recording and association with donor.
- Blood unit/component registration and life-cycle status.
- Inventory search, filtering and monitoring.
- Patient blood request creation and status management.
- Request review, approval/rejection and fulfillment tracking.
- Blood unit issue recording.
- Validation of core data and transaction states.
- Role-based access for major user categories.
- Audit history for key transactions.
- Operational dashboards and standard reports.

5.2 Out of Scope

- Cross Match Machine integration.
- Automated transfusion monitoring at bedside.
- Medical diagnosis or clinical decision-making.
- Payment, billing or insurance processing.
- Donor mobile application.
- External SMS/email provider implementation.

5.3 Constraints

- Workflows must align with hospital policy and applicable regulatory requirements.
- Only authorized staff based on access permission is to perform controlled actions.
- Production rules may differ by organization, jurisdiction and blood component.

6. Current-State (As-Is) Process:

Stage	Typical Activity	Risk
1. Donor Registration	Staff identifies/registers donor	Duplicate ,Incomplete,missing records
2. Donation	Donation details are captured	Manual tracking of donation history,maintain the records physically which leads to records being misplaced.
3. Collection	Blood units are collected and stored	Status may be difficult to track
4. Inventory	Eligible units are recorded for issue	Limited real-time visibility,incorrect stock status
5. Immuno Haematology	Serology test to be done for collected units	Results maybe interchanged with other donors records.
5. Request	Hospital staff raises requirements	Request status may be fragmented
6. Review	Staff checks availability and eligibility	Manual searching/checking
7. Issue	Unit is issued	Traceability may depend on manual records
8.Disposal	Disposal Of expired Units	Chances of disposing healthy Units
8. Reporting	Management compiles operational information	Time-consuming consolidation

7. Proposed Future-State (To-Be) Process

Step	Actor	System Behaviour
1. Register/Search Donor	Technician	Search existing donor; create or update record.
2. Record Donation	Technician	Capture donation event and generate/associate unit record.
3. Update Unit Status	Authorized staff	Move unit through configured life-cycle statuses.
4. Make Available	Authorized staff	Eligible unit becomes available inventory.
5. Create Request	Clinician/authorized staff	Enter patient, requirement and urgency information.
6. Review Request	Authorized reviewer	Validate request and check availability.
7. Allocate/Approve	Authorized reviewer	Reserve suitable eligible unit(s).
8. Issue	Technician	Confirm issue and update unit/request status.
9. Audit	System	Record user, timestamp, action and relevant status changes.

8. Functional Requirements

Area	Requirement	Priority
User authentication	System shall authenticate users before access to protected module functions.	Critical
Role-based access	System shall restrict functions according to configured user roles.	Critical
Donor search	Users shall be able to search donor records using permitted identifiers.	High
Donor registration	Users shall be able to create donor records with mandatory information.	High
Donor update	Authorized users shall be able to update permitted donor information.	High
Donation recording	Users shall be able to record a donation against a donor.	Critical
history	System shall display a donor's relevant donation history.	High
Unit creation	System shall create or register a unique blood-unit record.	Critical
Component capture	System shall record the blood component associated with a unit.	High
Blood group capture	System shall capture the applicable blood group information.	Critical
Unit status	System shall maintain a lifecycle status for each unit.	Critical
Eligibility control	System shall prevent issue of units that are not in an issue-eligible state.	Critical
Inventory search	Users shall filter inventory by blood group, component, status and other permitted criteria.	High
Expiry monitoring	System shall identify units approaching or past configured expiry.	High
Patient search	Authorized users shall search existing patient records where integrated/available.	High
Request creation	Authorized users shall create blood requests for a patient.	Critical
Request priority	System shall capture configured request urgency/priority.	High
Request status	System shall maintain request statuses such as Draft, Pending, Approved, Rejected, Fulfilled and Cancelled.	Critical

Request review	Authorized users shall review request details before approval/rejection.	Critical
Availability check	System shall display relevant eligible inventory during request processing.	Critical
Allocation	Authorized users shall allocate/reserve eligible unit(s) against a request.	Critical
Issue transaction	Authorized staff shall record the issue of allocated unit(s).	Critical
Issue history	System shall retain issue details including unit, request, user and timestamp.	Critical
Cancellation	Authorized users shall cancel eligible requests with a reason.	High
Rejection reason	System shall capture a reason for request rejection.	High
Audit trail	System shall record key create/update/status actions.	Critical
Dashboard	System shall display key inventory and request indicators.	Medium
Reports	Authorized users shall generate standard operational reports.	Medium
Export	Authorized users shall export permitted report data.	Medium
Validation	System shall validate mandatory fields and configured business rules.	Critical

9. Business Rules

Sl No	Business Rule
1	Only authenticated and authorized users may access controlled blood bank transactions.
2	A donor record should not be duplicated when an existing donor can be identified.
3	A blood unit must have a unique identifier within the system.
4	A unit cannot be issued unless its status is configured as eligible for issue.
5	Expired units must not be available for issue.
6	A request must identify the patient and required blood/component information before submission.
7	Rejected requests must contain a rejection reason.
8	Cancelled requests must not be fulfilled.
9	A completed issue transaction must update the relevant unit and request status.
10	Critical status changes must be attributable to an authenticated user and timestamp.
11	Users may only perform actions permitted by their role.
12	A unit reserved for one request must not be simultaneously allocated to another request unless explicitly released.
13	Mandatory fields and allowed values shall be configurable where business policy varies.
14	Inventory counts shall be derived from current unit status rather than manually entered totals.
15	Any exception to standard workflow must be captured with an appropriate reason and audit record.

10. Non-Functional Requirements

Sl No	Category	Requirement
1	Security	Access shall be controlled by authentication and role-based authorization.
2	Audibility	Critical transactions shall retain actor, timestamp, action and relevant before/after status where applicable.
3	Performance	Common searches and standard page operations should provide a responsive user experience under expected operational load.
4	Availability	The module should be available during defined hospital operating periods subject to planned maintenance.
5	Data Integrity	The system shall prevent invalid state transitions and inconsistent core records.
6	Usability	Common workflows should minimize unnecessary steps and clearly identify validation errors.
7	Privacy	Sensitive information shall only be accessible to users with a legitimate business need.
8	Scalability	The design should support growth in donors, units, requests and historical transactions.
9	Maintainability	Business rules and reference values that may change should be configurable where practical.
10	Recoverability	Data backup and recovery processes shall follow the host organization's IT policy.

10.1 Security Considerations

- Role-based authorization for create/update/approve/issue actions.
- Session management and secure authentication.
- Audit logs protected from unauthorized alteration.
- Least-privilege access to patient and donor information.
- Secure handling of exported reports.

11. Use Cases

Sl No	Use Case	Primary Actor	Summary	Expected Outcome
1	Register Donor	Technician	Create a new donor record after confirming no suitable existing record is present.	Donor record saved with mandatory fields.
2	Record Donation	Technician	Capture donation event and associate it with a donor.	Donation is saved and traceable.
3	Manage Blood Unit	Technician	Create/update unit information and lifecycle status.	Unit has unique identity and valid status.
4	Create Blood Request	Authorized staff	Raise a requirement for a patient.	Request enters Pending state.
5	Review Request	Authorized reviewer	Review request and determine next action.	Request is approved, rejected or returned according to policy.
6	Allocate Unit	Authorized reviewer	Select eligible inventory for an approved request.	Unit becomes Reserved/Allocated.
7	Issue Unit	Technician	Complete issue transaction.	Unit becomes Issued and request is updated.
8	Monitor Inventory	Administrator/Technician	Search and filter available inventory.	Relevant inventory is displayed.
9	Generate Report	Authorized user	Run operational report with filters.	Report is generated/exported according to permissions.

12. Data Requirements

Entity	Key Attributes	Relationship / Purpose
Donor	Donor ID, name, DOB/age, contact, eligibility-related fields	Parent record for donation history
Donation	Donation ID, donor ID, date/time, collection details, status	Links donor to collected blood
Blood Unit	Unit ID, donation ID, group, component, collection date, expiry, status	Core inventory item
Patient	Patient ID, demographics/identifiers permitted by system	Subject of blood request
Blood Request	Request ID, patient ID, group/component, quantity, priority, status, reason	Tracks demand

Allocation	Allocation ID, request ID, unit ID, timestamp, user	Links inventory to request
Issue	Issue ID, request ID, unit ID, timestamp, user, outcome	Records fulfilment
Audit Event	Event ID, entity, action, actor, timestamp, metadata	Traceability

12.1 Data Quality Requirements

- Unique identifiers shall be enforced for core entities.
- Required fields shall be validated before submission.
- Reference values such as status and component types should be centrally maintained.
- Historical transaction records should not be silently overwritten.
- Search results should clearly distinguish active and historical records.

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12.2 Conceptual Relationship

Donor 1 ──< Donation 1 ──< Blood Unit

Patient 1 ──< Blood Request 1 ──< Allocation >──1 Blood Unit

Blood Request 1 ──< Issue >──1 Blood Unit

13. Reporting & Dashboard Requirements

Report / KPI	Purpose	Key Filters
Available Inventory	Monitor current stock	Blood group, component, status
Low Stock	Identify shortage conditions	Blood group, component, threshold
Near Expiry	Identify units requiring attention	Expiry window, component
Pending Requests	Monitor unfulfilled demand	Priority, status, date
Request Turnaround	Measure operational performance	Date range, priority
Donation Summary	Monitor donations	Date range, donor/donation status
Issue Summary	Review fulfilment activity	Date range, component, group
Audit Report	Review controlled activity	User, entity, action, date

13.1 Dashboard KPIs

- Available units by blood group.
- Available units by component.
- Pending requests.
- High-priority requests.
- Units nearing expiry.
- Requests fulfilled versus pending.
- Donations recorded over selected period.

13.1 Alerts

- Alerting the Blood bank Technician about the High priority Emergency Request
- Units nearing expiry.
- Transfusion Reaction Alerts for a particular patient
- Alerts of Completion of Bleed while collecting a blood from a donor.

14. Future Enhancements

- Automated donor notification after a donation reaches an appropriate completed state.
- SMS/email notifications for selected request or inventory events.
- Predictive inventory and demand analytics.
- Advanced dashboards using BI tools.
- Blood bag,sample collection,cross match barcode/QR scanning.
- Integration with laboratory systems for automated cross matching,abo typing process.

- Integration with hospital EHR/EMR and patient master.
- Mobile-friendly workflows for authorized staff.

14.1 Portfolio Positioning

This case study demonstrates the ability to translate a healthcare operational problem into structured requirements, define stakeholders and workflows, document business rules, create testable acceptance criteria and establish traceability from business objectives to system behaviour.

15. Approval

Role	Name	Signature	Date
Business Owner			
Blood Bank Representative			
IT / Product Representative			
QA Representative			